Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of the **Intelligent Pharmacy Store Management System (IPSMS)**. The document is intended for developers, project supervisors, testers, and stakeholders to clearly understand the system's scope, features, and constraints.

The system aims to digitize and intelligently manage pharmacy operations while providing AI-assisted healthcare support to users.

1.2 Scope

The Intelligent Pharmacy Store Management System is a multi-role application that serves: - **Pharmacy Vendors** (store owners) - **Users / Patients** - **Doctors** (optional advisory role) - **Admin**

The system provides: - Medicine inventory management with batch-wise expiry tracking - AI-powered prescription reading and explanation - Smart medicine reminders for users - Stock prediction and expiry alerts for vendors - Secure authentication and role-based access control

The system will be accessible via a mobile application (Flutter) and optionally via a web dashboard.

1.3 Definitions, Acronyms, and Abbreviations

- **AI**: Artificial Intelligence
- **NLP**: Natural Language Processing
- **IPSMS**: Intelligent Pharmacy Store Management System
- **RLS**: Row Level Security
- **FEFO**: First Expiry First Out

1.4 References

- Gemini API Documentation
- PostgreSQL Documentation
- Supabase Authentication & Database Docs
- Flutter Framework Documentation

1.5 Overview

This document is structured into overall system description, functional requirements, non-functional requirements, external interface requirements, and system constraints.

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2. Overall Description

2.1 Product Perspective

IPSMS is a standalone system with cloud-based backend services. It integrates: - Flutter mobile frontend - Node.js backend APIs - PostgreSQL database - AI services for prescription understanding and recommendations

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2.2 User Classes and Characteristics

User Type	Description
Admin	Manages system, users, and overall monitoring
Vendor	Pharmacy owner managing inventory and sales
User	Customer purchasing medicines and uploading prescriptions
Doctor	Reviews prescriptions and provides optional guidance

2.3 Operating Environment

- Mobile OS: Android (primary), iOS (optional)
- Backend: Node.js with Express
- Database: PostgreSQL (Supabase)
- AI: Gemini API

2.4 Design and Implementation Constraints

- Medical data privacy must be maintained
- System must comply with basic healthcare data protection principles
- Internet connectivity is required for AI features

2.5 Assumptions and Dependencies

- Users possess smartphones with internet access
- Vendors correctly enter medicine batch data

- AI responses are advisory and not a replacement for doctors

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3. Functional Requirements

3.1 Authentication and Authorization

- Users shall be able to register and login using email and password
- System shall enforce role-based access control
- Vendors shall only access their own pharmacy data

3.2 Pharmacy Management (Vendor)

- Vendor shall be able to create and manage pharmacy profiles
- Vendor shall add, update, and remove medicines
- Vendor shall manage medicine batches with expiry dates
- System shall alert vendors about near-expiry and low-stock items

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3.3 Inventory Management

- System shall support multiple batches per medicine
- Stock shall be reduced using FEFO logic
- All stock movements shall be logged

3.4 Prescription Handling (User)

- User shall upload prescription images
- System shall extract text using AI
- System shall explain prescription in simple language
- System shall suggest medicine alternatives if unavailable

3.5 Orders and Sales

- User shall place orders from pharmacies
- System shall calculate total bill automatically
- Orders shall deduct stock from correct batches

3.6 Medicine Reminders

- User shall set medicine reminders
- System shall notify users based on schedule

- User shall activate or deactivate reminders

3.7 Admin Functions

- Admin shall manage users and pharmacies
- Admin shall monitor system activity

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4. Non-Functional Requirements

4.1 Performance Requirements

- System should handle concurrent users efficiently
- Inventory updates should reflect in real-time

4.2 Security Requirements

- Passwords must be securely hashed
- Sensitive data must be protected using access control
- Database must enforce row-level security

4.3 Reliability

- System should be available 99% of the time
- Data loss must be prevented via backups

4.4 Usability

- UI should be simple and intuitive
- Prescription explanations must be easy to understand

4.5 Scalability

- System should support multiple pharmacies
- Database design must support future AI features

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5. External Interface Requirements

5.1 User Interfaces

- Mobile UI built with Flutter
- Vendor dashboard with charts and alerts

5.2 Hardware Interfaces

- Smartphone camera for prescription upload

5.3 Software Interfaces

- Gemini API for AI processing
- Supabase for authentication and database

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6. System Models

6.1 Use Case Overview

- User Registration and Login
- Medicine Management
- Prescription Upload and Analysis
- Order Processing
- Reminder Scheduling

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7. Future Enhancements

- Online medicine delivery integration
- Payment gateway integration
- Advanced AI demand forecasting
- Doctor video consultation

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8. Conclusion

The Intelligent Pharmacy Store Management System provides a smart, scalable, and AI-driven solution to modern pharmacy challenges. By combining inventory intelligence with healthcare assistance, the system improves efficiency for pharmacies and safety for users.